

From Search to Synthesis: Implementing Multi-hop RAG

Moving Beyond "Single-Pass" Retrieval to
Complex Financial Reasoning

PRESENTED BY

DocCentralGPT

SPG-ML-LLM-DEV

DATE

December 2025

The Problem with Standard RAG in Finance

Why "Simple" Retrieval Fails the Analyst Test

The "Disconnect" Example:

USER QUERY

"Compare the EBITDA margins of Company A and its main competitor."

STANDARD RAG FAILURE

✗ Finds generic industry articles but misses the specific filings needed to calculate the margin. It cannot **"connect the dots"**.

What is Multi-hop RAG?

The "Planning-Reasoning" Architecture

Single-Hop vs. Multi-Hop

FEATURE	SINGLE-HOP (STANDARD)	MULTI-HOP (AGENTIC)
Query Type	Factual, explicit ("What is X?")	Complex, multi-part ("How does X impact Y?")
Data Source	Single Document / Chunk	Multiple Disconnected Documents
Process	Linear (One pass)	Iterative / Recursive (Looping)
Reasoning	Read and Summarize	Deduce and Connect
Fin. Example	"Show me the 2023 Annual Report."	"Is the CEO's compensation linked to ESG goals?"

STANDARD WORKFLOW: LINEAR



AGENTIC WORKFLOW: CYCLIC



Refines query based on intermediate results

The "Retrieve-Read-Reason" Loop



Query Decomposition

Breaking Down the "Deal Question"

KEY INSIGHT

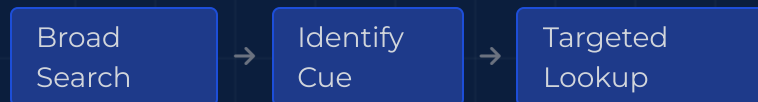
"You cannot answer Sub-Q 3 without knowing the answer to Sub-Q 1."

Recursive Retrieval

The "Drill-Down" Method

Instead of searching for specific details immediately, the system first retrieves a high-level summary to understand the context, then generates targeted queries for the details.

PROCESS LOGIC



Knowledge Graphs (GraphRAG)

Structured Reasoning for Corporate Hierarchies

❌ WHY VECTORS FAIL HERE

Vector databases find similar text, not connected entities. They struggle with explicit relationships like ownership, family trees, or supply chains.

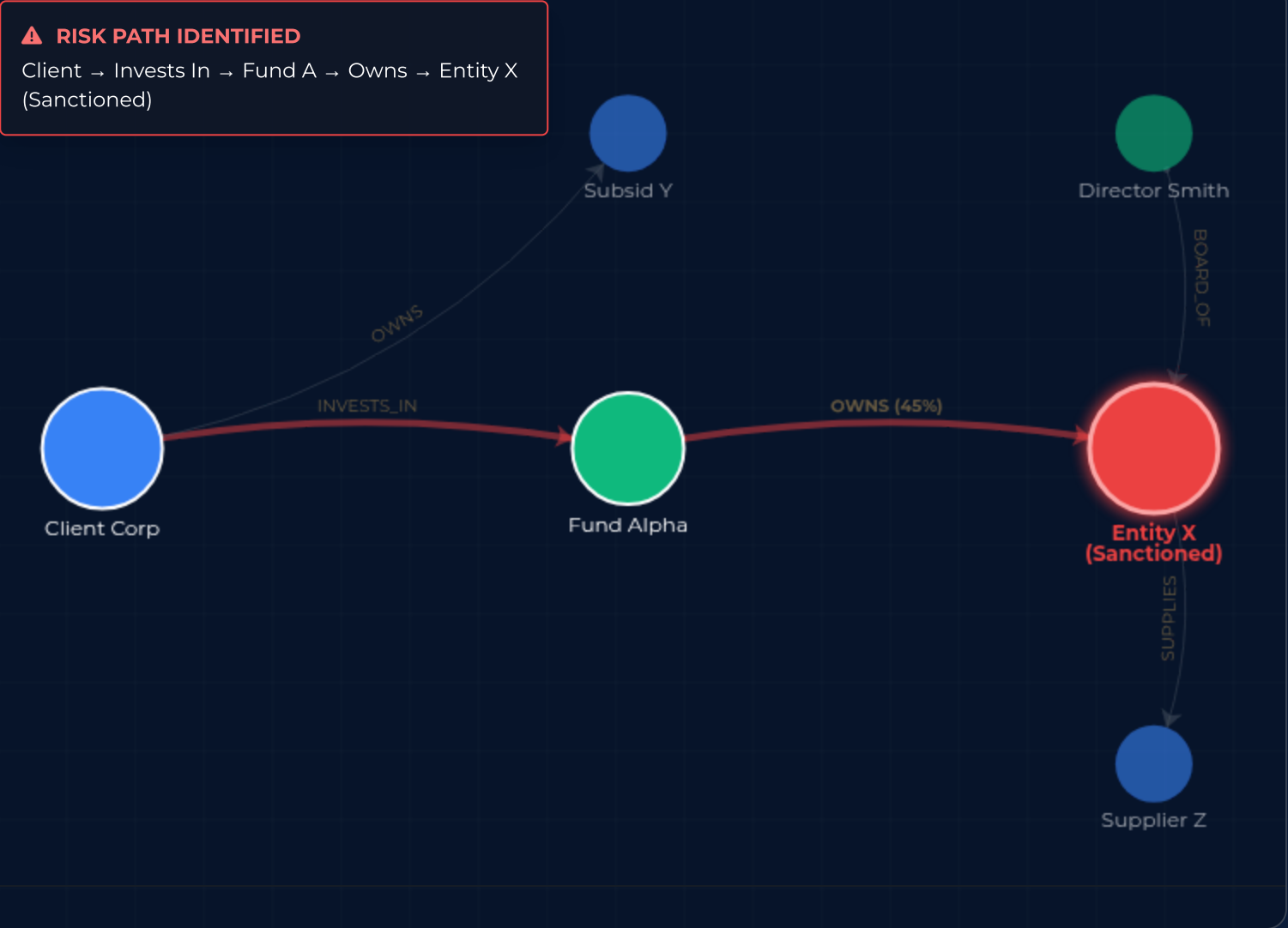
🔗 THE GRAPH ADVANTAGE

Data stored as **Nodes** (Companies, People) and **Edges** (OWNS, SUPPLIES).

The "Hop" = traversing graph edges, not text similarity.

GRAPH LEGEND

- Company
- Person/Fund
- Sanctioned
- Relationship



Ultimate Beneficial Owner (UBO)

Following the Ownership Trail

THE CHALLENGE

Scenario: Compliance check for a new investor.

"Does the new investor have any exposure to sanctioned entities?"

KEY INSIGHT

The "hop" follows relationship edges (A owns B, B invested in C) rather than text similarity. This path is often invisible to standard vector search which looks for keywords.

ENTITY TYPES

- Investor/Person
- Fund/Holding
- Company
- Sanction List



! RISK EXPOSURE DETECTED

Investor A → Fund B → Company C → OFAC

The Financial Bridge

The Reality: Siloed Data

Financial data lives in disconnected islands. Standard RAG finds one, but can't link them.



10-K
Filings



News/Web



Legal
Docs

✕ *No logical connection*

COMPLEX USER QUERY

"How will proposed tariff increases on Chinese EV imports impact the margins of Tesla's main battery supplier?"

The Multi-Hop Solution

 Connecting the Dots

Multi-Constraint Financial Query

The "Needle in a Haystack" Query

```
"Find all 2024 US SASB transactions secured  
by Hotels or Resorts that included  
condominium interest managed by the hotel but  
owned by third parties"
```

This single query contains four distinct constraints that typically reside in completely different sections of deal documentation or separate databases.

Three Hops to the Answer

1



Identify Univers

 SUB-QUESTION



M&A Valuation Comp (3 Hops)

Headline

Building Comparables from Scratch

You cannot search for an "average multiple" directly. The system must first identify the targets, retrieve their financials at deal time, and then compute the average.



Standard RAG fails to identify "Top 3 deals" dynamically

USER QUERY

"What was the average EV/EBITDA multiple of the three largest SaaS acquisitions in North America last quarter?"

Multi-Hop Execution Flow



Tracing Exposure Through the Value Chain

The Challenge

Tracing downstream risk requires connecting entity relationships, geographic revenue data, and regulatory events.



Supply Chain



Geo Rev



Regulations



Market Data

⚠ *Multi-domain reasoning
required*

COMPLEX USER QUERY

"Which of Tesla's battery suppliers has the highest revenue exposure to China, and how has their stock performed since US export controls were announced?"

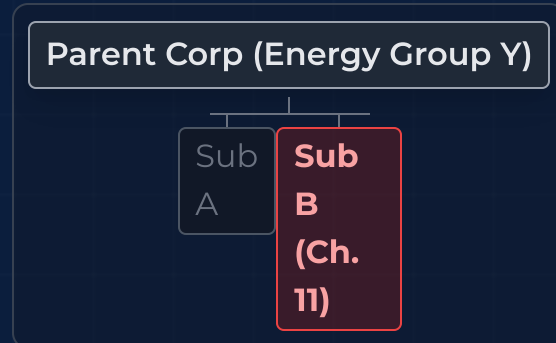
4-Hop Execution Path

 Chained Reasoning

Portfolio Contagion Check

The Hidden Risk

Bankruptcy contagion is often obscured by complex corporate structures. A subsidiary fails, but the exposure lies in the parent company's debt held by your funds.



⚠ *User doesn't know the Parent's name*

COMPLEX USER QUERY

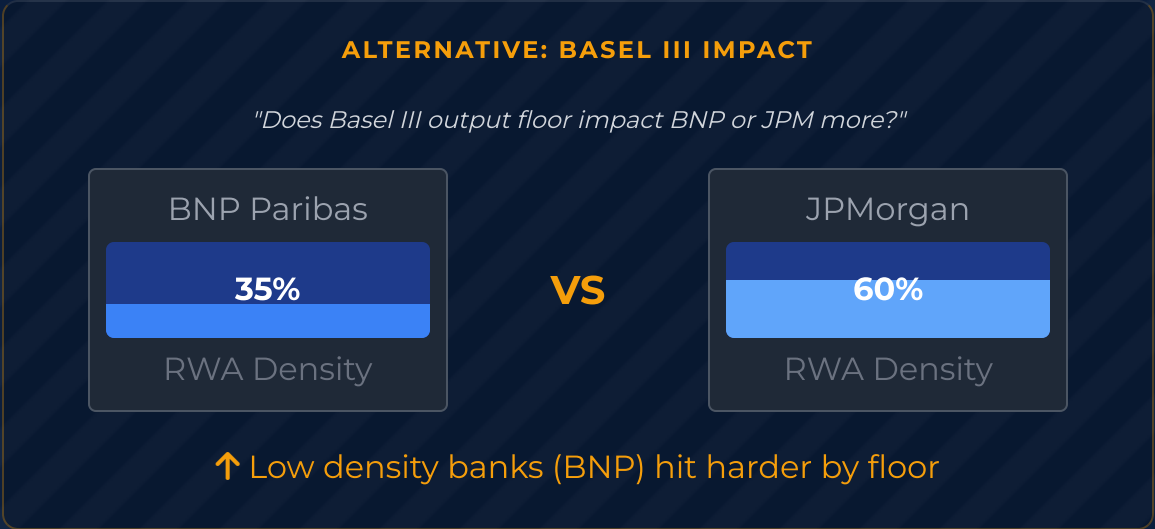
"Identify all mutual funds in our portfolio holding bonds from the parent company of the utility that filed Chapter 11 in Texas last week."

Traversing the Corporate Hierarchy

🔍 3-Hop Contagion Check

Use Case 4: Regulatory Arbitrage

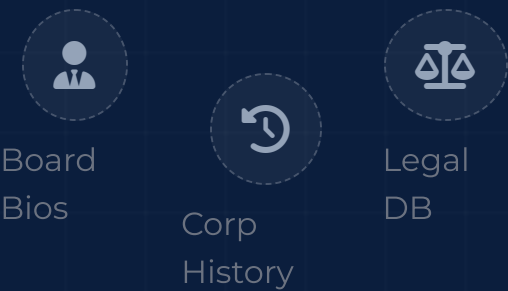
QUERY: "DOES UK OR NETHERLANDS OFFER BETTER TAX TREATMENT FOR IP ROYALTIES?"		
CRITERIA	👑 UK PATENT BOX (HOP 1)	🔄 NETHERLANDS INNOVATION BOX (HOP 2)
Tax Rate	10% Corporate Tax Rate	9% Effective Tax Rate
Qualifying Assets	Patents & Exclusive Licenses	Patents + R&D Software
Entry Requirement	Development condition	WBSO R&D Certificate
Synthesis	Higher rate, narrower scope	Advantageous for Software IP



Executive Red Flag Search

The Challenge

Standard background checks often miss indirect connections between new appointees and past companies with legal trouble.



✖ *Missing the "Risk Link"*

COMPLEX USER QUERY

"Do any board members appointed to Citi in the last 12 months have past directorships at companies currently under SEC investigation?"

The Multi-Hop Workflow

Automated Due Diligence

Timeline Reconstruction for Compliance

COMPLIANCE QUERY

"Did the CEO's stock sales occur before or after the announcement of the DOJ investigation?"

1 Hop 1: Event Anchor
Identify date of "DOJ investigation announcement"

2 Hop 2: Transaction Check
Retrieve dates of "CEO stock sales" from SEC Form 4

RECONSTRUCTED SEQUENCE

NOV 10

CEO Stock Sale
Sold 50k shares @ \$142
SEC Form 4

**RED FLAG DETECTED**



NOV 12

Investigation Public
DOJ announces probe
PR Newswire

WHY IT MATTERS

Precise sequencing identifies potential insider trading flags critical for regulatory reporting and litigation defense.

ANALYSIS RESULT:

Sales preceded announcement by 48 hours



Three Patterns of Multi-Hop



PATTERN 01

Causal Chain

SUPPLY CHAIN CONTAGION

EXAMPLE QUERY

"Which Apple supplier is most vulnerable to the Taiwan drought?"

- **Step 1: Entity List**
Retrieve list of Apple suppliers
- **Step 2: Location Mapping**
Map factory locations for each
- **Step 3: Event Overlay**
Cross-reference with drought zones



PATTERN 02

Comparative

REGULATORY ARBITRAGE

EXAMPLE QUERY

"Does the UK or Netherlands offer better IP tax treatment?"

- **Step 1: Retrieve A**
Get UK Patent Box regime details
- **Step 2: Retrieve B**
Get NL Innovation Box details
- **Step 3: Synthesis**
Compare qualifying criteria & rates



PATTERN 03

Timeline Check

TEMPORAL SEQUENCE

EXAMPLE QUERY

"Did CEO sales precede the investigation announcement?"

- **Step 1: Anchor Event**
Find date of investigation news
- **Step 2: Target Event**
Find dates of stock transactions
- **Step 3: Sequencing**
Compare timestamps (Delta calculation)

Three Patterns of Multi-Hop

Recognizing Query Types That Require Multi-Hop Reasoning



PATTERN 01

Inference Query
DOCUMENT COMPARISON

EXAMPLE QUERY

"Which report discusses the supply chain risk of Apple, the 2019 annual report or the 2020 annual report?"

HOP	SUB-QUESTION	SOURCE
Hop 1	Does Apple's 2019 10-K discuss supply chain risk?	Apple 10-K FY2019
Hop 2	Does Apple's 2020 10-K discuss supply chain risk?	Apple 10-K FY2020



SYNTHESIS

Compare findings → Determine which report contains the discussion



PATTERN 02

Comparison Query
CROSS-ENTITY METRICS

EXAMPLE QUERY

"Did Netflix or Google report higher revenue for the year 2023?"

HOP	SUB-QUESTION	SOURCE
Hop 1	What was Netflix's total revenue for FY2023?	Netflix 10-K FY2023
Hop 2	What was Alphabet's total revenue for FY2023?	Alphabet 10-K FY2023



SYNTHESIS

Compare revenue figures → Determine which is higher



PATTERN 03

Temporal Query
TIMELINE RECONSTRUCTION

EXAMPLE QUERY

"Did Apple introduce the AirTag before or after the 5th gen iPad Pro?"

HOP	SUB-QUESTION	SOURCE
Hop 1	When did Apple launch the AirTag?	Press releases, 8-K
Hop 2	When did Apple launch iPad Pro 5th gen?	Press releases, 8-K



SYNTHESIS

Compare dates → Determine temporal sequence: "AirTag launched [Date A]..."



Key Takeaway: All three patterns require 2 retrievals + 1 synthesis step. Single-hop RAG cannot answer these because the final answer requires connecting information across multiple sources.

Additional Multi-Hop Query Categories

i Beyond the core patterns of **Temporal**, **Comparison**, and **Inference**, financial analysis requires a broader range of multi-hop reasoning types.

#	QUERY TYPE	DESCRIPTION	FINANCIAL EXAMPLE
1	TEMPORAL	Compares timing or sequence of events	"Did X happen before or after Y?"
2	COMPARISON	Compares metrics across entities	"Which company has higher revenue?"
3	INFERENCE	Draws conclusions from document analysis	"Which report discusses topic X?"
4	BRIDGE	Finds entity in Doc A, uses it to query Doc B	"Who is the CEO of the company that acquired Slack, and what is their compensation?"
5	COMPOSITIONAL	Combines multiple facts to compute an answer	"What is the debt-to-equity ratio of the largest bank by assets?"
6	CAUSAL / IMPACT	Traces cause-effect relationships	"How did the Fed rate hike impact JPMorgan's NII?"
7	AGGREGATION	Collects & aggregates data across sources	"What is the average DSCR across all 2024 office CMBS deals?"
8	HIERARCHICAL / TRAVERSAL	Navigates ownership or corporate structures	"List all subsidiaries of Berkshire Hathaway exposed to flood insurance risk"
9	NULL / NEGATIVE	Confirms absence of information	"Did Tesla disclose any cybersecurity incidents in 2024?"
10	CONDITIONAL / FILTER	Applies multiple filtering criteria sequentially	"Find CLOs with reinvestment periods ending in 2025 that have CCC buckets above 7.5%"
11	COUNTERFACTUAL	Explores hypothetical scenarios vs actuals	"What would Apple's margin be if COGS increased 10%?"
12	VERIFICATION / FACT-CHECK	Cross-validates claims across sources	"Does the CEO's statement on the earnings call match the 10-Q disclosure?"

Basic Query Patterns

TEMPORAL QUERIES (SEQUENCE/TIMING)

#	QUERY	DOCUMENT TYPES	HOPS
1	"Did Apple introduce the AirTag before or after the launch of the 5th generation iPad Pro?"	Press Releases8-K Filings	2
2	"Was the Citrix leveraged loan amendment filed before or after Moody's downgraded the facility to Caa1?"	Credit AgreementRating Report	2
3	"Did SVB Financial disclose liquidity concerns in their 10-K before or after the stock price dropped 60%?"	10-K FilingMarket Data	2

COMPARISON QUERIES (CROSS-ENTITY METRICS)

#	QUERY	DOCUMENT TYPES	HOPS
4	"Did Netflix or Google report higher revenue for fiscal year 2023?"	10-K Filings (2)	2
5	"Which CLO manager had a higher AAA tranche subordination level in 2024: Carlyle or Apollo?"	CLO IndenturesPresale Reports	2
6	"Compare the debt service coverage ratio for the largest loan in BMARK 2024-V6 versus GSMS 2024-GC60"	CMBS Prospectus (2)	2

INFERENCE QUERIES (DOCUMENT CONTENT ANALYSIS)

#	QUERY	DOCUMENT TYPES	HOPS
7	"Which report discusses the supply chain risk of Apple, the 2019 annual report or the 2020 annual report?"	10-K Filings (2)	2
8	"Does Fitch or KBRA provide more detailed loss severity assumptions in their CMBS presale methodology?"	Rating Methodologies	2
9	"Which quarter's 10-Q first disclosed the material impact of chip shortages on Ford's production: Q1, Q2, or Q3 2021?"	10-Q Filings (3)	3

Entity & Calculation Patterns

BRIDGE QUERIES (ENTITY CHAINING)

#	QUERY	DOCUMENT TYPES	HOPS
10	"What is the credit rating of the parent company of the utility that filed Chapter 11 in Texas last month?"	News/8-KCorporate HierarchyRating Reports	3
11	"Who is the CFO of the company that acquired Slack, and what was their total compensation in 2023?"	M&A FilingsProxy Statement (DEF 14A)	3

COMPOSITIONAL QUERIES (MULTI-FACT CALCULATION)

#	QUERY	DOCUMENT TYPES	HOPS
13	"What is the EV/EBITDA multiple of the largest SaaS acquisition in North America in Q3 2024?"	M&A Announcements10-K/8-K Filings	3
14	"Calculate the weighted average coupon of the top 5 loans by notional in the CIFC CLO 2024-II portfolio"	CLO IndentureLoan Prospectuses	3

CAUSAL / IMPACT QUERIES (CAUSE-EFFECT ANALYSIS)

#	QUERY	DOCUMENT TYPES	HOPS
16	"How did the Boeing 737 MAX grounding impact Spirit AeroSystems' revenue concentration with Boeing?"	10-K Filings (pre/post)Earnings Transcripts	3
17	"What was the impact on Goldman Sachs' net interest income following the 2023 Fed rate hikes?"	10-K/10-QEarnings TranscriptsFed Announcements	3

Advanced Retrieval Patterns



AGGREGATION QUERIES (MULTI-DOCUMENT SUMMARIZATION)

#	QUERY	DOCUMENT TYPES	HOPS
19	"What is the average DSCR across all office CMBS deals issued in 2024 with loan balances over \$100M?"	CMBS Prospectus Supplements	4
20	"Calculate the median spread at issuance for all B-rated leveraged loans in the healthcare sector in H1 2024"	Loan Prospectuses LCD Data	3
21	"What is the total exposure to CCC-rated assets across all Blackstone-managed CLOs issued in 2023?"	CLO Trustee Reports	3



HIERARCHICAL / TRAVERSAL QUERIES (STRUCTURE NAVIGATION)

#	QUERY	DOCUMENT TYPES	HOPS
22	"List all subsidiaries of Berkshire Hathaway with exposure to California wildfire insurance risk"	10-K Exhibit 21 Subsidiary Filings Insurance Filings	3
23	"Which portfolio companies of KKR have borrowings in the leveraged loan market that are currently on watchlist?"	Fund Holdings Credit Agreements Rating Watchlists	4
24	"Trace the ownership chain from Investor X to any sanctioned entity through fund investments"	Investor Disclosures Fund Holdings OFAC List	4



CONDITIONAL / FILTER QUERIES (SEQUENTIAL FILTERING)

#	QUERY	DOCUMENT TYPES	HOPS
25	"Find all 2024 SASB CMBS deals secured by hotels with LTV > 65% and DSCR < 1.40"	CMBS Prospectus Supplements Presale Reports	4
26	"Identify leveraged loans issued in 2024 with sponsor equity below 30% and no financial maintenance covenants"	Loan Prospectuses Credit Agreements	3
27	"List CLOs with reinvestment periods ending in 2025 that currently have CCC bucket utilization above 7.5%"	CLO Indentures Trustee Reports	3

Validation & Edge Case Patterns

○ NULL / NEGATIVE QUERIES (ABSENCE CONFIRMATION)

#	QUERY	DOCUMENT TYPES	HOPS
28	"Did Nvidia disclose any material weakness in internal controls in their 2024 10-K?"	10-K Filing	2
29	"Has Tesla ever disclosed a cybersecurity incident in any SEC filing between 2020-2024?"	10-K, 8-K Filings (5 years)	3
30	"Did the CLO indenture for Ares XLVII include an ESG exclusion clause for thermal coal investments?"	CLO Indenture	2

💬 COUNTERFACTUAL QUERIES (HYPOTHETICAL SCENARIOS)

#	QUERY	DOCUMENT TYPES	HOPS
31	"What would Tesla's gross margin be if lithium prices increased 20% from current levels?"	10-K (COGS breakdown) Commodity Data	3
32	"How would JPMorgan's CET1 ratio change if commercial real estate losses doubled from current reserves?"	10-K Basel III Disclosures	3
33	"What would the DSCR be for the Marriott CMBS loan if RevPAR declined 15% from underwritten levels?"	CMBS Prospectus Appraisal	3

✓ VERIFICATION / FACT-CHECK QUERIES (CROSS-SOURCE VALIDATION)

#	QUERY	DOCUMENT TYPES	HOPS
34	"Does the EBITDA figure in the company's investor presentation match the adjusted EBITDA in the audited 10-K?"	Investor Presentation 10-K Filing	2
35	"Does Moody's reported WARF for CLO 2024-X match the trustee's calculated WARF in the monthly report?"	Moody's Surveillance Trustee Report	2
36	"Does the CEO's statement on job growth in the earnings call align with headcount disclosed in the 10-K?"	Earnings Transcript 10-K Filing	2

💡 **Key Insight:** Single-hop RAG handles **lookup**. Multi-hop RAG handles **analysis**. The query type determines the retrieval strategy.

Key Challenges to Watch For



Latency Overhead

IMPACT: 3-5X SLOWDOWN

Every hop adds an LLM inference call and database retrieval cycle. A standard query takes <2s, while a 3-hop chain can exceed 10s.

- Wait time increases linearly with hops
- User experience degrades for real-time needs

⚡ MITIGATION STRATEGY

Parallelize retrieval where branches are independent. Implement aggressive caching for intermediate steps.



Context Noise

RISK: "LOST IN THE MIDDLE"

Accumulating > 100 documents across multiple hops creates a noisy context window. The LLM may overlook early critical clues or hallucinate connections.

- Performance drops as context length grows
- Irrelevant details from Hop 1 pollute Hop 3

⚡ MITIGATION STRATEGY

Summarize intermediate results before passing to next hop. Use "Chain-of-Note" to filter noise at each step.



Error Propagation

RISK: CASCADING FAILURE

Errors cascade. If Hop 1 identifies the wrong "Target Company", all subsequent financial retrievals will be for the wrong entity.

- Early hallucination ruins entire chain
- Harder to debug than single-shot errors

🛡️ MITIGATION STRATEGY

Implement validation steps ("Is this the correct company?"). Use confidence scoring to abort low-quality paths early.

Strategic Value & ROI



TIME TO INSIGHT

Significant reduction in
research time

"Hours of work in minutes"



ACCURACY

Minimized hallucination rate
on facts

"Grounded in specific filings"



AUDITABILITY

Fully traceable decision logic

"Every hop is logged"



SCALABILITY

Increased analyst throughput
capacity

"Handle cross-team queries"

The Operational Shift

TRANSFORMATION



Standard RAG

LEGACY APPROACH

- Simple Q&A over single documents
- Analyst must manually connect dots
- Context limited to keyword similarity



Multi-Hop Agent

TARGET STATE



- ✓ Intelligent multi-step reasoning across corpus
- ✓ System autonomously synthesizes answers
- ✓ Deep context understanding via knowledge graph

Conclusion

Multi-Hop RAG = The Analyst's Force Multiplier

